KVL derivation for three-phase unbalanced system

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Introduction

In this document, I review two papers. The first papers introduces the derivation is for three-phase, balanced systems. The second derviation is for three-phase unbalanced systems.

First Paper: Network Reconfiguration in Distribution Systems for Loss Reduction and Demand Balancing, Baran et. al.

Paper's Objectives

1- How can we change the network configuration to reduce resistive losses? 2- How can the load be distributed such that no overloads occur on the T-Lines?

Equations Derivation

NOTE: The paper uses DistFlow equations, which we are going to use in this work.

Consider the following single line diagram:

For each load, we consider P_{L_n} and Q_{L_n}, indicating load power flow at each load.

Each line has an impedance, $Z_n = r_n + jx_n$. The apparent power flow at each line is given by:

$$\bar{S_n} = P_n + jQ_n$$

For instance, at bus $i+1$:

$$\bar{S}_{i+1} = P_{i+1} + jQ_{i+1}$$

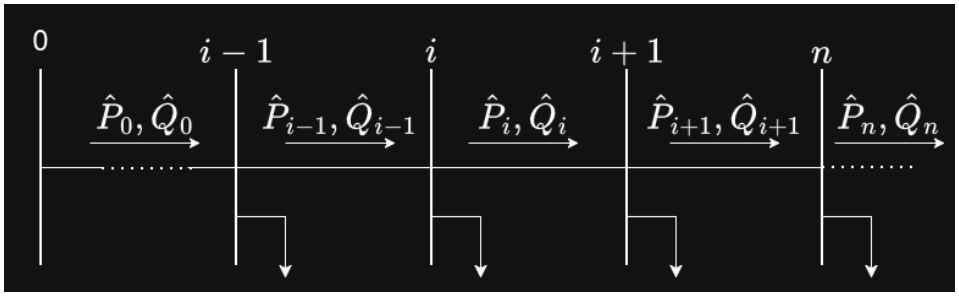


Figure 1: A single Line Diagram (SLD) for an n bus system. The loads are unknown for now. The Apparent Power ($\bar{S}$) flowing between each bus is shown.

I think it is easier to see this if we look at the power flow between two buses. Consider the power flow between buses 0 & $i-1$:

$$\underbrace{\bar{P}_0}_{\text{Power leaving bus 0}} = \underbrace{\bar{P}_{0,loss}}_{\text{Losses}} + \underbrace{\bar{P}_{i-1}}_{\text{Power entering bus } i-1} + \underbrace{\bar{P}_{i-1,load}}_{\text{Power consumed by bus } i-1 \text{ load}} \tag{1}$$

Decomposing the Losses term in equation Equation 1

$$P_{losses} = r|I^2|$$

At any given bus n:

$$P_{n,losses} = r_n|I_n^2| \tag{2}$$

Recall:

$$\bar{S} = \bar{I}^*\bar{V} = \bar{P} + j\bar{Q}$$

Wherein the magnitude of the apparent power is given by:

$$|S| = |I|\ |V| = \sqrt{P^2 + Q^2}$$

Squaring both sides:

$$|S|^2 = |I|^2\ |V|^2 = P^2 + Q^2$$

Solving for the current, I:

$$|I|^2 = \frac{P^2 + Q^2}{|V|^2}$$

At any given bus n, the current is:

$$|I_n|^2 = \frac{P_n^2 + Q_n^2}{|V_n|^2} \tag{3}$$

substituting the current equation Equation 3 into Equation 2 we get:

$$P_{n,losses} = r_n \frac{P_n^2 + Q_n^2}{|V_n|^2} \tag{4}$$

Subtituting Equation 4 in the KCL between buses 0 and $i-1$ in Equation 1, we get:

$$\underbrace{\bar{P}_n}_{\text{Power leaving bus } n} = \underbrace{r_n \frac{P_n^2 + Q_n^2}{|V_n|^2}}_{\text{Losses}} + \underbrace{\bar{P}_{n+1}}_{\text{Power entering bus } n+1} + \underbrace{\bar{P}_{n+1,load}}_{\text{Power consumed by bus } n+1 \text{ load}} \tag{5}$$

> **Note**
>
> Equation 5 is the real power DistFlow Equation. This equation can be used for any power flow calculations.

Similarly, the reactive power can be calculated in a similar manner:

$$\underbrace{\bar{Q}_n}_{\text{Q leaving bus } n} = \underbrace{x_n \frac{P_n^2 + Q_n^2}{|V_n|^2}}_{\text{Losses}} + \underbrace{\bar{Q}_{n+1}}_{\text{Q entering bus } n+1} + \underbrace{\bar{Q}_{n+1,load}}_{\text{Q consumed by bus } n+1 \text{ load}} \tag{6}$$

Deriving the Voltage Equation:

Applying KVL to buses, i and $i-1$ in Figure 1:

$$\bar{V}_{i-1} = \bar{Z}_{i-1}\ \bar{I}_{i-1} + \bar{V}_i \tag{7}$$

The current complex value can be solved for in a similar manner as in equation Equation 3:

$$\bar{V}_n\ \bar{I}_n^* = \bar{P}_n + j\bar{Q}_n$$

Solving for $\bar{I}_n^*$:

$$\bar{I}_n = \frac{\bar{P}_n - j\bar{Q}_n}{\bar{V}_n^*} \tag{8}$$

Subtituting Equation 8 in Equation 7 & solving for $\bar{V}_i$:

$$\bar{V}_i = \bar{V}_{i-1} - \bar{Z}_{i-1}\ \bar{I}_{i-1} \tag{9}$$

$$\bar{V}_i = \bar{V}_{i-1} - \bar{Z}_{i-1}\ \frac{\bar{P}_{i-1} - j\bar{Q}_{i-1}}{\bar{V}_{i-1}^*}$$

We can then extend this to get only the magnitude of the voltage:

Recall $|a^2 - b^2| = (a-b)(a^* - b^*)$:

$$|V_i^2| = (\bar{V}_{i-1} - \bar{Z}_{i-1}\ \bar{I}_{i-1})\ (\bar{V}_{i-1}^* - \bar{Z}_{i-1}^*\ \bar{I}_{i-1}^*)$$

Multiplying both terms, we end up with:

$$|V_i^2| = |V_{i-1}^2| - 2\ \Re\{\underbrace{\bar{V}_{i-1}^*\ \bar{I}_{i-1}}_{\substack{\text{Apparent}\\ \text{Power}\\ \text{Conjugate}}}\ \bar{Z}_{i-1}\} + |Z_{i-1}^2|\ |I_{i-1}^2| \tag{10}$$

Recall $\bar{S}_{i-1}^* = \bar{V}_{i-1}^*\ \bar{I}_{i-1} = \bar{P}_{i-1} - j\bar{Q}_{i-1}$

So the middle term can be decomposed such that:

$$\bar{V}_{i-1}^*\ \bar{I}_{i-1}\ \bar{Z}_{i-1} = (\bar{P}_{i-1} - j\bar{Q}_{i-1})(r_{i-1} + jx_{i-1})$$

Multiplying and obtaining only the real part because we are trying to calculate the magnitude:

$$\Re\{\bar{Z}_{i-1}\ \bar{V}_{i-1}^*\ \bar{I}_{i-1}\} = r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1} \tag{11}$$

The last term in Equation 10 can be simplified further. Recall that $|Z_{i-1}^2| = r_{i-1}^2 + x_{i-1}^2$ and $|I_{i-1}^2| = \frac{P_{i-1}^2 - Q_{i-1}^2}{|V_{i-1}^2|}$

$$|Z_{i-1}^2|\ |I_{i-1}^2| = (r_{i-1}^2 + x_{i-1}^2)\ (\frac{P_{i-1}^2 - Q_{i-1}^2}{|V_{i-1}^2|}) \tag{12}$$

Subtituting Equation 11 and Equation 12 in Equation 10:

$$|V_i^2| = |V_{i-1}^2| - 2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1}) + (r_{i-1}^2 + x_{i-1}^2)\ \frac{P_{i-1}^2 - Q_{i-1}^2}{|V_{i-1}|^2} \tag{13}$$

> **Note**
>
> Equation 13 is the Voltage Magnitude DistFlow Equation. This equation can be used for any $n - bus$ power system.

Three-phase Unbalanced System

Using the same KVL we used before, Equation 9, we can set a system of equations for buses i and j such that:

$$\begin{bmatrix} \bar{V}_i^a \\ \bar{V}_i^b \\ \bar{V}_i^c \end{bmatrix} = \begin{bmatrix} \bar{V}_j^a \\ \bar{V}_j^b \\ \bar{V}_j^c \end{bmatrix} - \begin{bmatrix} \bar{Z}_{ij}^{aa} & \bar{Z}_{ij}^{ab} & \bar{Z}_{ij}^{ac} \\ \bar{Z}_{ij}^{ba} & \bar{Z}_{ij}^{ab} & \bar{Z}_{ij}^{bc} \\ \bar{Z}_{ij}^{ca} & \bar{Z}_{ij}^{cb} & \bar{Z}_{ij}^{cc} \end{bmatrix} \begin{bmatrix} \bar{I}_{ij}^a \\ \bar{I}_{ij}^b \\ \bar{I}_{ij}^c \end{bmatrix}$$

For any given bus, the voltage equation can extracted from the above matrix form. For instance:

$$\bar{V}_i^a = \bar{V}_j^a - \bar{Z}_{aa}\ \bar{I}_{ij}^a - \bar{Z}_{ab}\ \bar{I}_{ij}^b - \bar{Z}_{ac}\ \bar{I}_{ij}^c$$